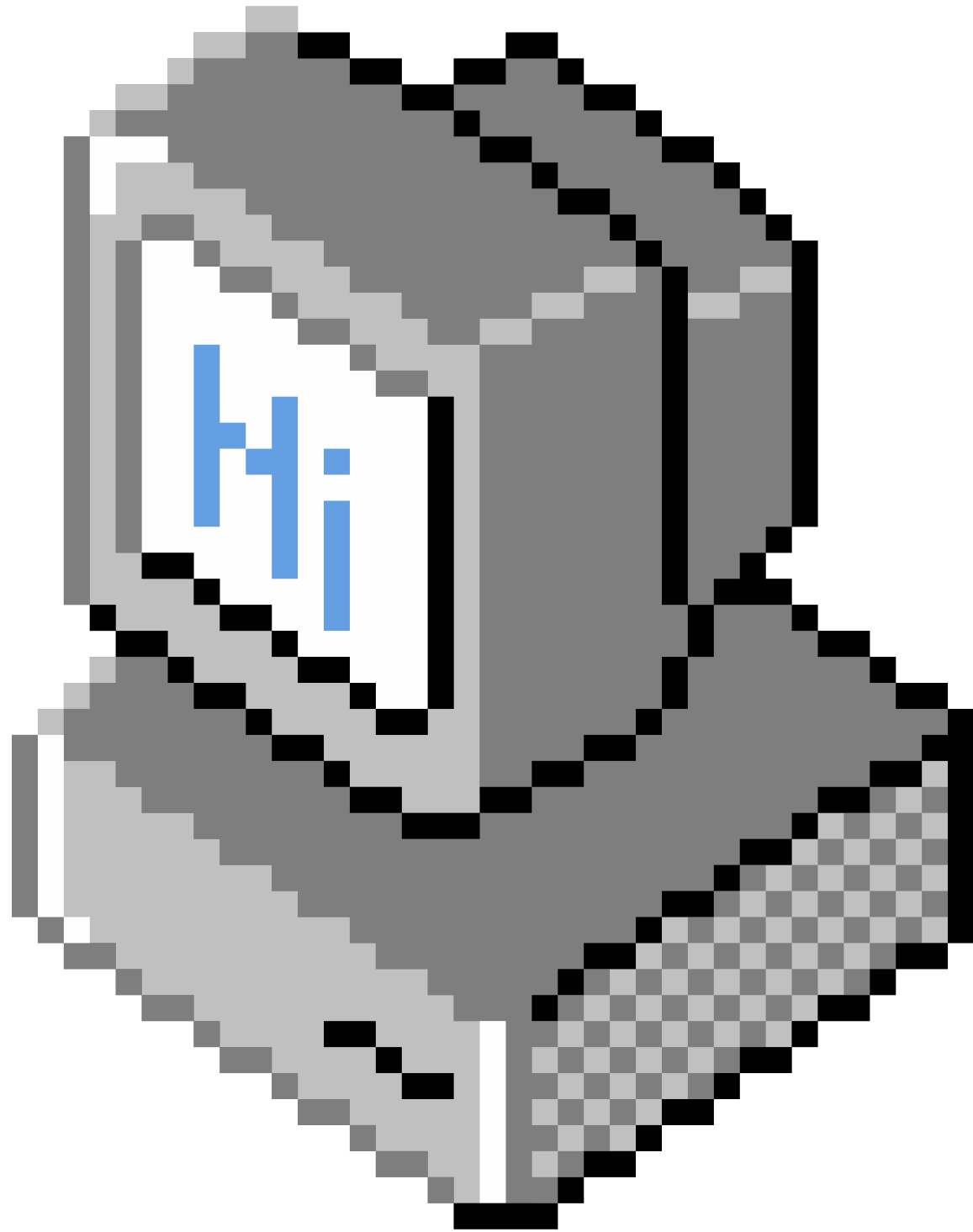




OSINT AND DIGITAL FORENSICS

**Understanding OSINT, DFIR, geolocation,
metadata, forensics, OPSEC.**

**HOW MUCH CAN THE INTERNET
REVEAL ABOUT YOU**



LET'S TEST IT. SEARCH THIS UP

A Google search bar with a rounded rectangular shape. On the left is the Google 'G' logo. The search text is "<your_surname_name> Aissms MhtCET". On the right is a button labeled "AI Mode" with a right-pointing arrow.

 <your_surname_name> Aissms MhtCET AI Mode →

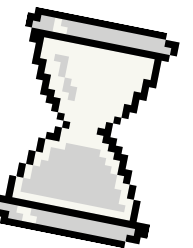
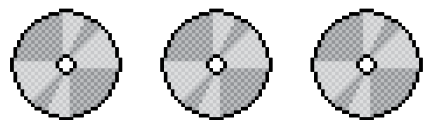
For example : **koul sushant aissms mhtcet**

WHAT IS INTELLIGENCE?



Intelligence is information that has been **collected**, **checked** and **analysed** so that it helps someone make a decision.

Raw facts alone are not intelligence – they become intelligence only **when someone understands what they mean** and what to do about them.



V · T · E				Intelligence management	[hide]
Collection	Human (HUMINT)	Special reconnaissance (organizations)			
		Clandestine	Asset recruiting · Cell system · Covert action · Direct action · Operational techniques		
		Espionage	Agents (field · handling) · Asset · Black operation (black bag) · Concealment device · Cover · Cryptography · Cutout · Dead drop · Defection / Turncoat · Denial and deception · Eavesdropping · False flag · Industrial espionage · Interrogation · Numbers station · One-way voice link · Resident spy · Steganography · Surveillance		
	Signals (SIGINT)	By alliances, nations and industries · In modern history · Operational platforms by nation · Direction finding · Traffic analysis · TEMPEST			
	Measurement and signature (MASINT)	Electro-optical · Geophysical · Nuclear · Radar · Radiofrequency · Materials · Casualty estimation (earthquake)			
	Other	Financial (FININT) · Geospatial (GEOINT) · Imagery (IMINT) · Open-source (OSINT) · Technical (TECHINT)			
Analysis	Competing hypotheses · Target-centric · Words of estimative probability · Intelligence source and information reliability · All-source intelligence · Basic intelligence · Intelligence assessment · Medical intelligence · Military geography · Scientific & Technical intelligence				
Dissemination	Intelligence cycle security · Counterintelligence (organizations) · Counterintelligence and counter-terrorism organizations				

WHAT IS OSINT?

OSINT stands for **Open Source Intelligence**. It means **collecting information** that is **freely** and **legally** available to the public and turning it into something useful.

"Open source" here does not mean open-source software. It means **open, public sources** – anything anyone can see **without hacking or breaking in**.

OPEN SOURCE

Search engines

Google, Bing, DuckDuckGo

Social media

Instagram, LinkedIn, X (Twitter), Facebook, YouTube

Public records

Company registrations, court records, government websites

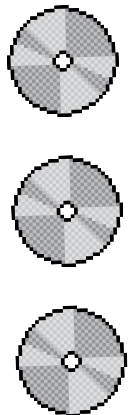
Websites and domains

WHOIS records, old versions of sites (Wayback Machine)

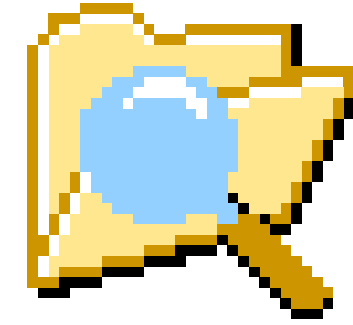
Images and maps

Photos, Google Maps, Google Earth, satellite images

News, forums, blogs



WHAT IS DIGITAL FORENSICS?



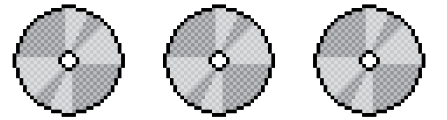
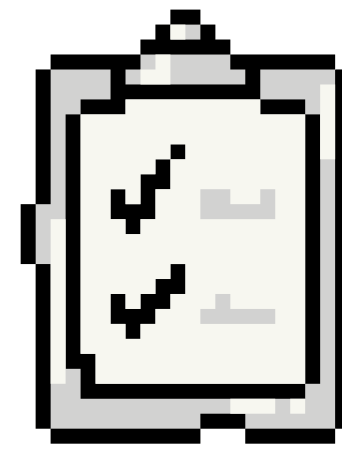
Digital forensics means **finding, collecting and studying evidence** from digital devices – computers, phones, hard disks, USB drives, cloud accounts and networks – **in a way that can be trusted in court.**

Simple example: A company's data is stolen. A forensic expert makes an exact copy of the employee's laptop, finds deleted files, checks browser history and USB logs, and **proves the data was copied to a pen drive on a certain date.**

OSINT VS DIGITAL FORENSICS

	OSINT	Digital Forensics
Data source	Public information on the internet	Seized devices, disks, logs, memory
Access needed	None – anyone can see it	Permission, warrant or company authority
Main question	What can we find out from public data?	What exactly happened on this device?
Example	Finding where a photo was taken	Recovering deleted chats from a phone

OSINT IN CYBER & IRL



CYBER-

Penetration testing (recon)

Before testing a company, ethical hackers collect public info – employee names, email formats, subdomains, technologies used. This is the first step of any attack, so defenders must know it too

Threat intelligence

Tracking hacker groups, new malware and attack plans discussed on forums and the dark web

Attack surface check

Finding forgotten servers, open ports and exposed devices using tools like Shodan

Phishing defence

Checking what employee details are public, because attackers use them to write believable fake emails

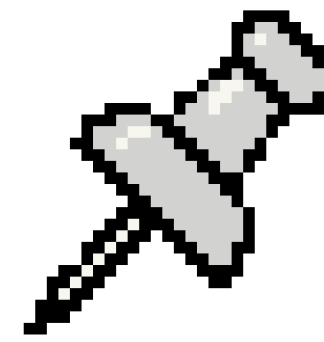
Incident investigation

Finding who is behind an attack using usernames, IPs and domains

IRL-

- **Journalism**
- **Police and law enforcement**
- **Business**
- **Personal safety**
- **Human rights**

COMMON OSINT TOOLS



Tool	What it does	Link
Google Dorks	Special Google search operators to find exact information (list of dorks in the Google Hacking Database)	www.exploit-db.com/google-hacking-database
theHarvester	Collects emails, subdomains and names for a domain	github.com/laramies/theHarvester
Sherlock	Checks which websites a username exists on	github.com/sherlock-project/sherlock
Have I Been Pwned	Checks if an email appeared in a known data breach	haveibeenpwned.com

QUIZZ 1 :-



- 1. Is looking at someone's public Instagram profile OSINT?**
- 2. When you delete a file, what usually stays on the disk?**

ANSWERS

1. Yes, if you only look at public data
2. The data, until something overwrites it

OSINT METHODOLOGY - THE INTELLIGENCE CYCLE

1. Planning

Decide the goal. What exactly do you need to find, and why? Is it legal?

2. Collection

Gather data from public sources – search engines, social media, maps, records

3. Processing

Clean and organise the data. Remove duplicates, translate, save screenshots with date and link

4. Analysis

Connect the dots. What does the data really show? Check it against more than one source

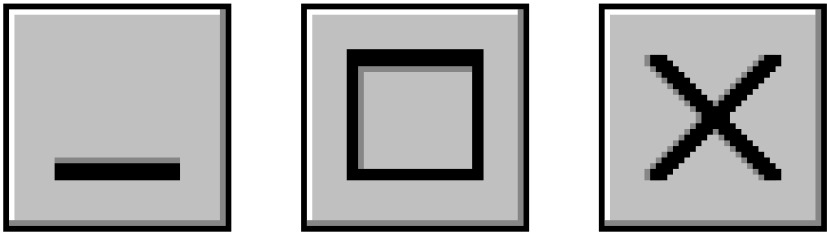
5. Reporting

Write clear findings with proof (links, screenshots, timestamps)

6. Feedback

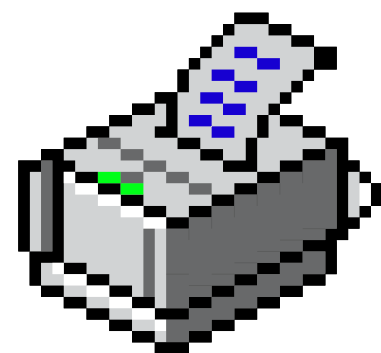
New questions come up, so the cycle starts again

PASSIVE VS ACTIVE OSINT



	Passive	Active
Meaning	Only look at public data, never touch the target	Interact with the target directly
Target knows?	No	May notice
Example	Reading someone's public LinkedIn, Google search	Port scanning their server, sending a friend request

INTRO TO DFIR



DFIR stands for Digital Forensics and Incident Response. It is what a company does when it gets hacked.

EXAMPLE >>>>> A THIEF BREAKS INTO A SHOP

- First, the police and guards rush in, catch or chase out the thief, lock the doors and make sure the shop can open again. That is Incident Response – act fast and stop the damage
- Then a detective comes in, carefully collects fingerprints and CCTV footage, and finds out how the thief got in, what was stolen and who did it. That is Digital Forensics – investigate and collect proof

In DFIR, the "shop" is a computer or network, the "thief" is a hacker, and the "fingerprints" are things like log files, deleted files and network traffic.

DFIR



- **Incident Response (IR)** – "Stop it!": act fast when an attack is happening – stop it, remove the hacker and get systems running again
- **Digital Forensics (DF)** – "What happened?": investigate carefully – how the hacker got in, what they did, what they took, and collect evidence

Real example: a company's computers suddenly show a ransomware message. The IR team unplugs the infected PCs from the network so it cannot spread, and restores files from backups. The forensics team then finds the phishing email that started it, sees which files were stolen, and writes a report for the police and management.

METHODOLOGY



Tool	Used for
Autopsy	Free tool to examine disk images, find deleted files and build timelines
FTK Imager	Makes exact forensic copies (images) of drives
Volatility	Analyses RAM (memory) dumps
Wireshark	Analyses network traffic captures
ExifTool	Reads file and photo metadata

- 1. Identification
- 2. Preservation
- 3. Collection
- 4. Examination & Analysis
- 5. Reporting

QUIZZ 2: PICK THE ANSWER

Finding the phishing email that started a ransomware attack is:

- A) Incident Response
- B) Forensics
- C) Passive OSINT

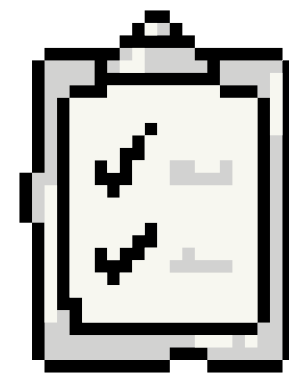
QUIZZ 2: PICK THE ANSWER

Finding the phishing email that started a ransomware attack is:

- A) Incident Response
- B) Forensics
- C) Passive OSINT

ANSWER B) Forensics

REVERSE IMAGE SEARCH

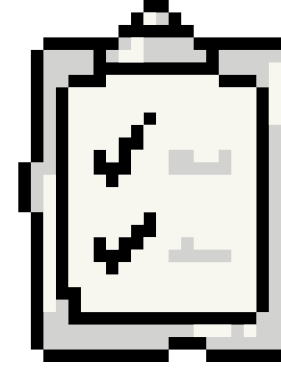


Normally you type words to find images. Reverse image search works the other way: you upload a picture and it finds where else that picture (or similar ones) appears online.

Uses:

- Find the original source and first date of a photo
- Catch fake news – an "earthquake today" photo that is actually 5 years old
- Catch fake profiles – a dating or scam profile using someone else's photos
- Identify a place, building, product or landmark in a photo

REVERSE IMAGE SEARCH TOOLS



Tool	Best for	Link
Google Lens / Google Images	Objects, products, landmarks, general search	lens.google.com
Yandex Images	Very good at finding faces, buildings and places	yandex.com/images
Bing Visual Search	Good second opinion, similar images	www.bing.com/visualsearch
TinEye	Finding exact copies and the oldest version of an image (sort by "Oldest")	tineye.com

REVERSE IMAGE SEARCH ADVANCED TOOLS

Tool	What it does	Key tip
Picarta (picarta.ai)	AI guesses where a photo was taken: country, city, GPS and a confidence score	A guess, not proof – confirm with Google Earth or Street View
Google Earth	3D globe with Street View and historical imagery	Match buildings and terrain; use historical imagery (Earth Pro) to date a photo

GEOSINT



GEOSINT (Geospatial OSINT) means **finding the exact place where a photo or video was taken**, using only clues visible in it.

Clue	What it can tell you
Language and signs	Country or region; shop names can be searched directly
Number plates	Country and sometimes state (shape, colour, format)
Driving side	Left side (India, UK, Japan) or right side (USA, Europe)
Road markings, poles, bollards	Each country has its own style

METHODOLOGY

1. Check the metadata first – the photo may already have GPS
2. Run a reverse image search – the photo may already be online with a location
3. Write down every clue in the image
4. Narrow down: continent → country → city → street
5. Use Google Maps, Street View and Google Earth to find the exact spot
6. Confirm by matching at least 2–3 features (a building + a pole + a mountain in the back)

Practice: the game GeoGuessr trains exactly this skill.

Safety lesson: this is why you should be careful posting photos from your home, school or with your house number visible.

EXAMPLE :

WHAT DOES IT SAY HERE :

**51.48449044020304,
0.08799148642372842**





HINT :
USE GOOGLE EARTH

ANSWER :



Ground Truth Verification or Street-Level Signage Analysis

TAKE HOME PROBLEM :



WATCH THE SOLUTION :
[YOUTUBE](#)

DIGITAL FORENSICS

Digital forensics is like **detective work for computers and phones**. Every action on a device leaves small traces – **opening a file, visiting a website, plugging in a USB, sending a message**. A forensic investigator finds these traces and uses them to tell the story of **what happened, when, and who did it**.

Simple rule: in digital forensics, **nothing is ever fully gone**. Even deleted files, cleared history and wiped chats often leave traces behind.

DIGITAL FORENSICS

Type	What it looks at	Example
Computer / Disk forensics	Hard disks, SSDs, USB drives	Recovering deleted files from a laptop
Mobile forensics	Phones and tablets	Getting WhatsApp chats, call logs, photos and location history
Network forensics	Network traffic and logs	Finding which IP stole data using Wireshark captures
Memory (RAM) forensics	What was running in memory	Finding malware that never saved itself to disk
Email forensics	Email headers and content	Proving a phishing email's real sender
Cloud forensics	Data in Google Drive, AWS, Microsoft 365	Checking who downloaded files from a company drive
Malware forensics	Harmful programs	Studying ransomware to see what it does

WHAT EVIDENCE CAN BE FOUND?



- **Deleted files:** when you delete a file, usually only its "address" is removed – the data stays on the disk until something overwrites it
- **Browser history, downloads and cookies:** which websites were visited and when
- **Log files:** logins, errors, system events – like a diary the computer keeps
- **USB history:** which pen drives were plugged in and when
- **Recently opened files and programs:** Windows keeps lists of these (e.g. in the Registry and Prefetch)
- **Storage:** Chats, emails, call logs and photos on phones
- **Metadata:** hidden details inside files

IMPORTANT TERMINOLOGIES FOR BEGINNERS

Term	Simple meaning
Evidence	Any data that helps prove what happened
Forensic image	An exact, bit-by-bit copy of a whole disk – including deleted and empty space. Investigators always work on this copy, never on the original
Write blocker	A device that lets you read a disk but blocks any writing, so the evidence cannot be changed by mistake
Hash	A digital fingerprint (like SHA-256). If the hash of the copy matches the original, the copy is exact and untouched
Chain of custody	A record of every person who handled the evidence and when
File carving	Recovering deleted files by searching the raw disk for known file patterns (e.g. every JPG starts with the same bytes)
Timeline	Putting all events in time order to see the full story
MAC times	Modified, Accessed and Created dates of a file

POPULAR FORENSIC TOOLS



Tool	Used for	Link
Autopsy	Free, easy tool to examine disk images, recover deleted files and build timelines	www.autopsy.com
FTK Imager	Makes forensic images of disks and memory	www.exterro.com/digital-forensics-software/ftk-imager
Volatility	Analyses RAM (memory) dumps	volatilityfoundation.org
Wireshark	Network forensics - analyses captured traffic	www.wireshark.org
ExifTool	Reads and removes file metadata	exiftool.org

METADATA AND ITS IMPORTANCE

Metadata is "data about data". It is hidden information stored inside a file that describes the file itself. Simple example: a photo shows a beach. That is the data. But hidden inside the file is: which phone took it, the exact date and time, and the GPS location. That is the metadata.

File type	Metadata it can contain
Photos (EXIF)	Camera / phone model, date and time, GPS location, camera settings, software used to edit
Documents (Word, PDF)	Author name, company name, created and modified dates, editing time, software version, printer
Emails (headers)	Sender's mail servers, IP addresses, time it was sent, the path it travelled
Audio / Video	Device, date, length, sometimes GPS
Every file (file system)	MAC times – Modified, Accessed, Created dates

WHY METADATA IS IMPORTANT

- **Forensic evidence:** proves when a file was created or changed and on which device – helps build a timeline of events
- **Catching fakes:** a photo claimed to be "taken today" but its metadata says 2019, or it shows Photoshop was used
- **Finding location:** GPS in a photo shows exactly where it was taken
- **OSINT recon:** documents on a company website can leak employee names, usernames and software versions that attackers can target
- **Privacy risk:** you may share your home location just by sending a photo



REAL CASES



Case 1: BTK Killer (USA, 2005)

- Who: Dennis Rader, a serial killer who murdered 10 people in Wichita, Kansas between 1974 and 1991. He named himself "BTK" and was not caught for about 30 years
- His mistake: in 2004 he started sending letters to police and the media again. He asked police if a floppy disk could be traced – they said no
- The clue: in 2005 he sent a floppy disk to a TV station. Investigators found a deleted Word file on it. Its metadata showed the organisation "Christ Lutheran Church" and the user "Dennis"
- The result: a quick internet search showed Dennis Rader was the church council president. DNA confirmed it, he was arrested in February 2005 and got 10 life sentences
- Lesson: deleted does not mean gone, and documents carry hidden author details



REAL CASES

Case 2: John McAfee (2012)

- Who: John McAfee, founder of the McAfee antivirus company
- What happened: in November 2012, police in Belize wanted to question him about his neighbour's death. He went into hiding and kept posting online about being on the run
- The clue: journalists from Vice magazine travelled with him and posted a photo of him taken on an iPhone. They forgot to remove the EXIF metadata – the GPS coordinates showed he was in Guatemala
- The result: his location became public within hours, and he was arrested in Guatemala a few days later for entering the country illegally
- Lesson: always remove metadata before posting photos online

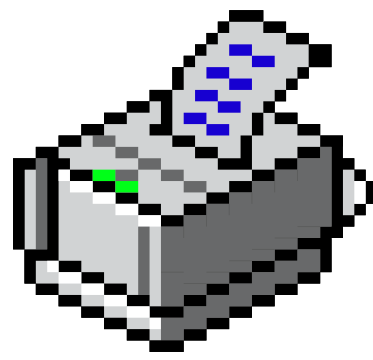




QUIZ 3: ANSWER IT YOURSELF

What should you do to a photo before posting it online?

EXIFT00L



EXIF (Exchangeable Image File Format) is the standard for the metadata that cameras and phones save inside photos.

ExifTool is a free command-line tool that can read, edit and remove metadata from almost any file – photos, videos, PDFs, Word files and more.

SETUP(In your codespaces)

```
cd challenges/Protocol_Day_3  
sh setup_day3.sh
```

Repo: github.com/omsapate23/protocol-cybersecurity-101-bootcamp

USEFUL COMMANDS

Command	What it does
<code>exiftool photo.jpg</code>	Shows all metadata of the photo
<code>exiftool -gps:all photo.jpg</code>	Shows only the GPS (location) data
<code>exiftool -n -gpslatitude -gpslongitude photo.jpg</code>	GPS as plain numbers – paste into Google Maps
<code>exiftool -make -model photo.jpg</code>	Which camera / phone took it
<code>exiftool -datetimeoriginal photo.jpg</code>	When the photo was actually taken
<code>exiftool -software photo.jpg</code>	Which software saved it (e.g. Photoshop = edited)
<code>exiftool -r folder/</code>	Read every file inside a folder
<code>exiftool -all= photo.jpg</code>	Removes all metadata (keeps a backup called photo.jpg_original)
<code>exiftool -b -ThumbnailImage photo.jpg > thumb.jpg</code>	Saves the small hidden thumbnail inside the photo

LEVEL 1: WHERE WAS THIS TAKEN?



Go to Challenges; then Protocol_Day_3 on your codespace

File: tuscan_photo.jpg

```
exiftool tuscan_photo.jpg
```

```
exiftool -make -model tuscan_photo.jpg
```

```
exiftool -gpslatitude -gpslongitude -n tuscan_photo.jpg
```

QUESTIONS

- 1. Which camera took it?**
- 2. Which country is it?**

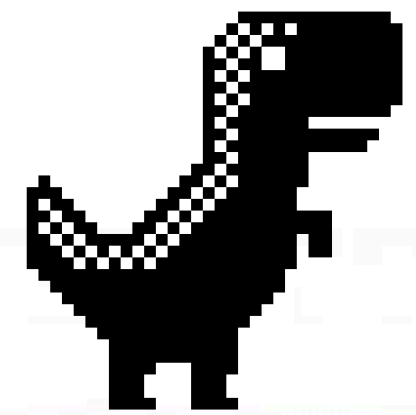
IMAGE FORENSICS

Image forensics means checking if a photo is real, edited or fake, and finding hidden information inside it.

What to check:

1. **Metadata:** date, device, GPS and Software field. Missing metadata can also be a clue (it was removed or the image was re-saved)
2. **Thumbnail mismatch:** the hidden thumbnail inside the photo may still show the original image before it was edited
3. **Error Level Analysis (ELA):** re-saves the image and shows which parts compress differently. Pasted or edited areas often glow brighter than the rest
4. **Clone detection:** finds parts of the image that were copied and pasted (e.g. extra smoke or crowds)
5. **Light and shadows:** shadows pointing in different directions mean objects were added
6. **Reverse image search:** find the original, unedited version online
7. **Hidden data:** files hidden inside the image (binwalk) or secret messages (steganography tools like steghide)

IMAGE FORENSICS



File type	Metadata it can contain
Photos (EXIF)	Camera / phone model, date and time, GPS location, camera settings, software used to edit
Documents (Word, PDF)	Author name, company name, created and modified dates, editing time, software version, printer
Emails (headers)	Sender's mail servers, IP addresses, time it was sent, the path it travelled
Audio / Video	Device, date, length, sometimes GPS
Every file (file system)	MAC times – Modified, Accessed, Created dates

LEVEL 2: SOMETHING IS HIDDEN.

GO TO YOUR CODESPACE CHALLENGES AND THEN TO Protocol_Day_3

File: hidden_photo.jpg

COMMANDS:

```
exiftool hidden_photo.jpg  
binwalk hidden_photo.jpg  
binwalk -e hidden_photo.jpg  
cat _hidden_photo.jpg.extracted/flag.txt
```

QUESTIONS

1. What does binwalk find after the JPEG?

2. What is the flag?

AUDIO FORENSICS

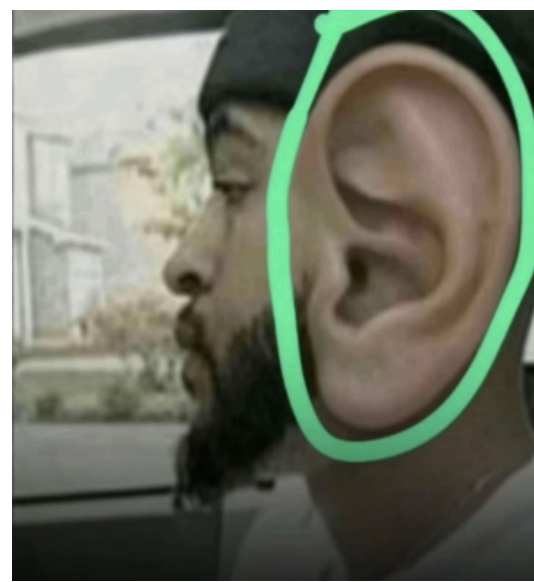
Audio forensics means **studying sound recordings to find the truth** – whether a recording is real or edited, who is speaking, what is being said, and when or where it was recorded. Simple example: someone gives the police a voice recording as proof. An audio forensic expert checks if any words were **cut**, **added** or **joined**, cleans up the noise so the speech is clear, and checks if the voice really belongs to the person.

Where It Is Used

- **Court cases:** proving a recording is original and not edited
- **Fake voice (deepfake) detection:** AI can now copy anyone's voice, so experts check if a voice is real
- **Enhancement:** making unclear speech easier to hear (removing hiss, traffic, wind)
- **Speaker identification:** comparing a voice with known samples of a person
- **CTF challenges:** hidden messages are often hidden inside audio files

AUDIO FORENSICS

Hidden Data in Audio (CTF)



- **Spectrogram images:** text or pictures drawn into the sound that only appear in the spectrogram view
- **Morse code:** beeps of short and long sounds
- **DTMF tones:** phone keypad tones that can be decoded into numbers
- **Steganography:** secret files hidden inside WAV files (tools: steghide, DeepSound)
- **Reversed or slowed audio:** play it backwards or at a different speed

Useful Commands :

```
exiftool recording.mp3          # read audio metadata
ffmpeg -i recording.mp3         # show file details
sox recording.wav -n spectrogram -o spec.png  # save a spectrogram picture
steghide extract -sf secret.wav  # try to extract hidden data
```

LEVEL 3: SEE THE SOUND.

GO TO YOUR CODESPACE CHALLENGES AND THEN TO PROTOCOL_DAY_3

File: secret_spectrogram.wav

COMMANDS:

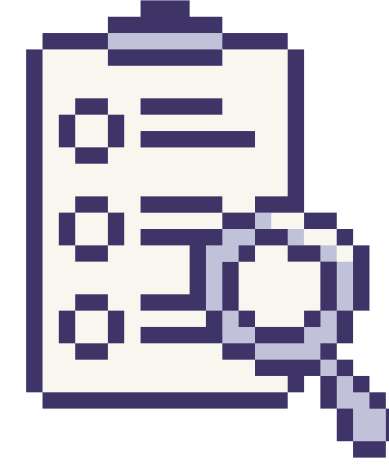
```
sox secret_spectrogram.wav -n spectrogram -o spec.png
```

Open spec.png

QUESTIONS

1. What flag appears in the picture?

OPSEC (OPERATIONS SECURITY)



OPSEC means **protecting your own information** so that others cannot use it against you. OSINT is about finding information on others – **OPSEC** is about not leaking information about yourself.

The idea came from the **US military** during the **Vietnam War**, when the enemy **kept learning about attacks in advance** – not from spies, but from **small public details that added up**.

Simple example: you post "**Flying to Goa for a week!**" with a photo of your **boarding pass**. Now strangers know your **house is empty**, and the **barcode on the pass can reveal your booking details**. That is **bad OPSEC**.

OPSEC (OPERATIONS SECURITY)



The 5 Steps of OPSEC:

1. **Identify critical information:** what must stay secret? (home address, passwords, travel plans, company projects)
2. **Analyse threats:** who might want it? (hackers, stalkers, scammers, competitors)
3. **Analyse vulnerabilities:** how could they get it? (social media posts, photo metadata, weak passwords)
4. **Assess the risk:** how likely is it, and how bad would it be?
5. **Apply countermeasures:** fix it – change settings, remove posts, use better tools

OPSEC (OPERATIONS SECURITY)



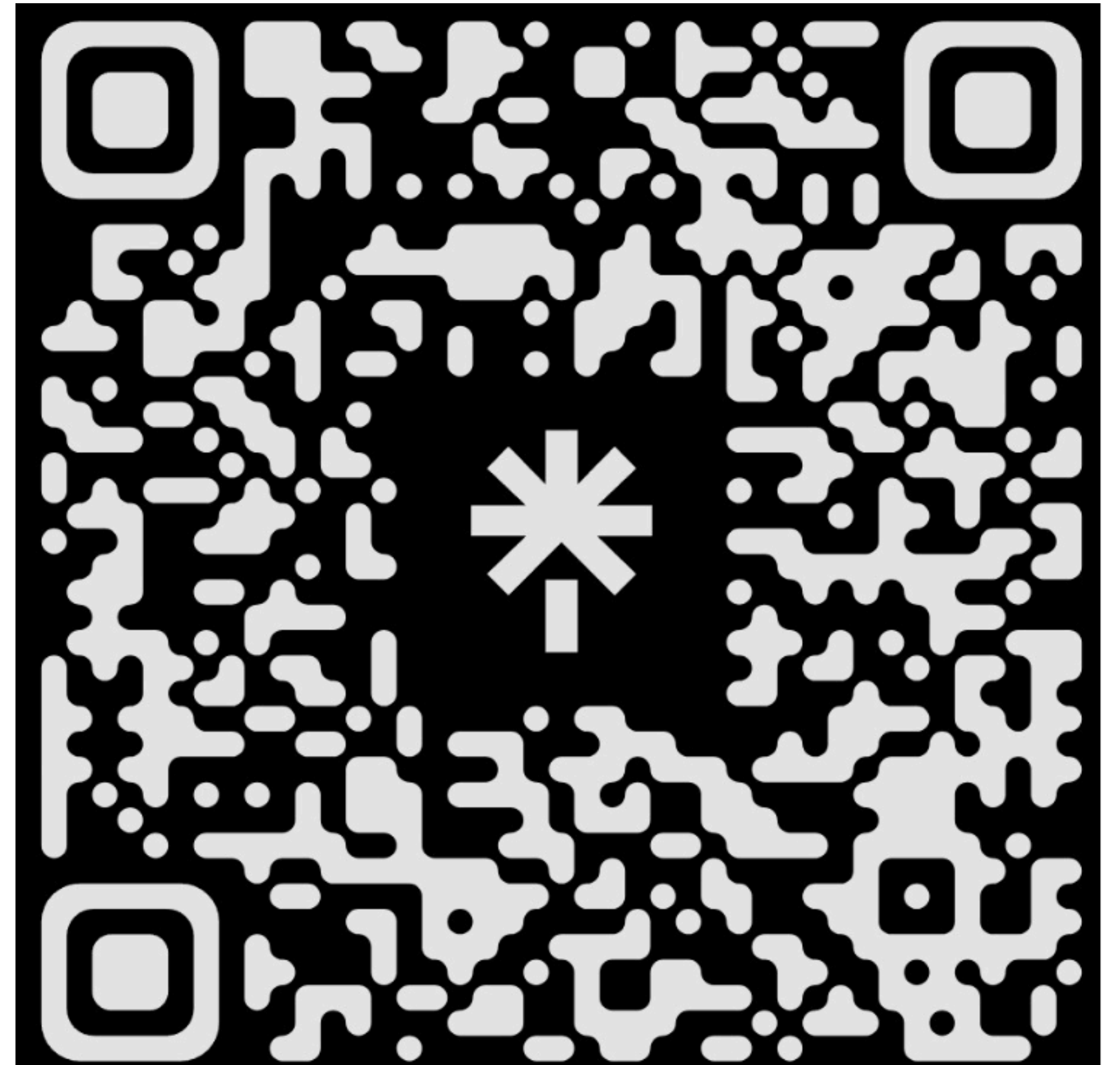
Real Example: **Strava Heatmap (2017–2018)**

The fitness app Strava published a **global map** showing where its users went running.

Soldiers using **fitness trackers** at **secret military bases** had their running routes on the map – **so the locations and layouts of military bases became visible to anyone.**

No hacking was needed. It is a famous lesson that even harmless data can **break OPSEC.**

THANK YOU



JOIN OUR COMMUNITY



**JOIN
THE
CTF**